Explainable Financial Alerts for Retail Investors: Integrating Statistical Anomaly Detection and News–Market Impact Correlation

Henrique José da Silva Santos

Student No.: 1180934

A dissertation submitted in partial fulfillment of the requirements for the degree of Master of Science, Specialisation Area of

Supervisor: Luís Gomes
Co-Supervisor: Rafael Silva

Evaluation Committee:
President:
[A definir]

Members:
[A definir]

Porto, June 21, 2026

Statement Of Integrity

I hereby declare that I have conducted this academic work with integrity.

I have not plagiarised or applied any form of undue use of information or falsification of results along the process leading to its elaboration.

Therefore, the work presented in this document is original and was authored by me, having not been previously used for any other purpose.

I declare that I have fully acknowledged the Code of Ethical Conduct of P.PORTO.

Use of Artificial Intelligence. In accordance with the P.PORTO/ISEP guidelines, I declare that artificial intelligence tools (notably Claude Code) were used during the development of this work to assist with the drafting and editing of the text and with software development. I directed the work and critically reviewed and validated all outputs; the ideas, decisions and final content are my own and are my responsibility.

ISEP, Porto, June 21, 2026

Dedicatory

Abstract

This dissertation presents an explainable (XAI-first) financial-alert system for retail investors in the US equity market (NYSE and NASDAQ). The system raises Telegram alerts from two independent triggers — abrupt market movements, detected as statistical anomalies, and incoming financial news — and, for every alert, exposes a fully traceable explanation: the detected event, the reasoning applied, the sources, and historical precedents. Its core is a news–market correlation engine that, given a news item, retrieves analogous historical news from a knowledge base built on the FNSPID dataset and measures the impact those precedents had on prices. As an Artificial Intelligence Engineering contribution, the work integrates, applies and critically evaluates existing components in a functional, explainable and reproducible system, with a documented news-impact correlation methodology.

Keywords: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

Resumo

Esta dissertação apresenta um sistema de alertas financeiros explicável (XAI-first) para investidores de retalho no mercado acionista dos Estados Unidos (NYSE e NASDAQ). O sistema envia alertas via Telegram a partir de dois gatilhos independentes — movimentos abruptos de mercado, detetados como anomalias estatísticas, e novas notícias financeiras — e, para cada alerta, expõe uma explicação totalmente rastreável: o evento detetado, o raciocínio aplicado, as fontes e os precedentes históricos. O seu núcleo é um motor de correlação notícia–mercado que, dada uma notícia, recupera notícias históricas análogas de uma base de conhecimento construída sobre o conjunto de dados FNSPID e mede o impacto que esses precedentes tiveram nos preços. Enquanto contribuição de Engenharia de Inteligência Artificial, o trabalho integra, aplica e avalia criticamente componentes existentes num sistema funcional, explicável e reprodutível, com uma metodologia documentada de correlação notícia–impacto.

Palavras-chave: Explainable AI, Financial Alerts, Anomaly Detection, News Impact, Retail Investors, Financial NLP

Acknowledgement

Contents

List of Figures

List of Tables

List of Algorithms

List of Source Code

List of Symbols

r_t	log-return of an asset on day t	—
μ_t	rolling mean of returns	—
σ_t	rolling standard deviation of returns	—
z_t	z-score of the return on day t	—
k	anomaly threshold (in standard deviations)	—
w	rolling-window length	d

List of Acronyms

AI	Artificial Intelligence.
NASDAQ	National Association of Securities Dealers Automated Quotations.
NYSE	New York Stock Exchange.
SIFMA	Securities Industry and Financial Markets Association.
XAI	Explainable Artificial Intelligence.

Chapter 1

Introduction

This chapter introduces the dissertation. It motivates the work, states the problem, defines the objectives and research questions, summarises the scientific contributions, and outlines the structure of the document.

1.1 Motivation and Overview

The United States equity market is the largest in the world and is owned, directly or indirectly, by a majority of US households: 62% of Americans reported owning stock in 2025 (Gallup 2025), in a market whose capitalisation reached US$62.2 trillion at the end of 2024 (Securities Industry and Financial Markets Association (SIFMA) 2025). At the same time, artificial intelligence has been adopted rapidly across financial services (Cambridge Centre for Alternative Finance 2026), yet many of the resulting models behave as opaque "black boxes" whose reasoning users cannot inspect (Barredo Arrieta et al. 2020). Chapter 2 develops this context in detail.

For the non-professional ("retail") investor, this combination is difficult. Markets move continuously across thousands of tickers and financial news arrives without pause, but the tools that help to interpret these signals are typically institutional, opaque, or both. This dissertation addresses that gap with an *explainable* financial-alert system: one that not only flags relevant events but also explains them in a way the investor can follow, verify and act upon.

1.2 Problem Statement

A retail investor who wishes to stay informed faces two distinct problems. First, distinguishing a genuinely abnormal price movement from ordinary volatility requires statistical judgement and constant attention. Second, when a relevant news item appears, assessing its likely impact requires knowledge of how analogous past events affected prices — information that is rarely at hand. Existing consumer tools tend to deliver alerts without explanation, or rely on models whose internal logic is hidden, which is precisely the transparency problem that explainable AI seeks to address (Adadi and Berrada 2018; Barredo Arrieta et al. 2020).

The problem this work tackles is therefore not price prediction, but *explanation*: detecting abrupt market movements and relevant news, and presenting, for each alert, a traceable account of what was detected, the reasoning applied, the sources, and concrete historical precedents of similar events and their observed impact.

1.3 Objectives and Research Questions

The general objective is to design, implement and evaluate an explainable, reproducible financial-alert system for US retail investors, driven by two independent triggers (abrupt market movements and incoming news) and delivered through Telegram. This is pursued through three research questions:

- **RQ1.** Can transparent statistical methods detect abrupt market movements with enough precision to be useful to a retail investor, while remaining explainable?
- **RQ2.** Can a news–market correlation engine retrieve genuinely analogous historical news and quantify their observed price impact, without lookahead, better than trivial baselines?
- **RQ3.** Can the resulting explanations be made faithful to the system's actual logic and useful to a retail investor?

The supporting objectives are: (i) build a thin end-to-end slice early; (ii) develop each component in its simplest defensible form; (iii) evaluate each component against a design defined beforehand (Chapter 6); and (iv) keep the whole system reproducible and transparent.

1.4 Scientific Contributions

This is an Artificial Intelligence *Engineering* dissertation. The contribution is not to invent new algorithms but to *integrate, apply and critically evaluate* existing components in a functional, explainable and reproducible system. Concretely, the contributions are:

- A documented, reproducible methodology for **news–market impact correlation**, combining sentence embeddings for retrieval with event-study windows for impact measurement, with explicit choices of similarity metric and window and explicit avoidance of lookahead.
- An **XAI-first alerting pipeline** that exposes its full reasoning chain (detection, evidence, sources, precedents) to a non-expert user.
- A **critical evaluation** of each component against baselines and a transparent design, reporting honest results and limitations.

1.5 Document Structure

The remainder of the document is organised as follows. Chapter 2 sets the market and technological context. Chapter 3 reviews the relevant literature on anomaly detection, explainable AI and financial NLP, with comparative tables. Chapter 4 describes the system architecture and the method chosen for each component. Chapter 5 details the implementation. Chapter 6 reports the evaluation. Chapter 7 concludes with the main findings, limitations and future work.

Chapter 2

Contextualization

This chapter sets the context for the work. It outlines the scale of the United States equity market, the growth of retail investing, the rapid adoption of Artificial Intelligence (AI) in finance and the explainability concerns it raises, and the information-overload problem that motivates an explainable financial-alert system.

2.1 The US Equity Market (NYSE and NASDAQ)

The United States equity market is the largest and among the deepest and most liquid in the world. According to the Securities Industry and Financial Markets Association (SIFMA) Capital Markets Fact Book, US equity markets represented 49.1% of the US$126.7 trillion in global equity market capitalisation at the end of 2024 — about US$62.2 trillion, some 5.3 times the size of the next-largest market, China (Securities Industry and Financial Markets Association (SIFMA) 2025). The two principal trading venues are the New York Stock Exchange (NYSE) and National Association of Securities Dealers Automated Quotations (NASDAQ), which together list the large-capitalisation companies that dominate global indices.

This market has also grown substantially over the last decade. As shown in Figure 2.1, US equity market capitalisation rose from US$25.1 trillion in 2015 to US$62.2 trillion in 2024 (Securities Industry and Financial Markets Association (SIFMA) 2025), a scale and liquidity that make it the natural focus for a retail-oriented alerting system.

2.2 The Retail-Investing Landscape

Equity ownership is widespread among US households. In 2025, 62% of Americans reported owning stock — whether individual shares or holdings within mutual funds and retirement accounts — matching the 2024 figure and continuing a rebound after more than a decade below that level (Gallup 2025). Ownership is, however, unevenly distributed: it rises to 87% among households earning US$100,000 or more and falls to 28% among those earning under US$50,000 (Gallup 2025).

Beyond ownership, individual ("retail") investors have become a material part of day-to-day trading activity, helped by commission-free mobile brokerages. Published estimates of the retail share of US equity trading volume vary widely with methodology and period — broadly in the range of 20–35% during 2024–2025. This combination of broad ownership and growing direct participation defines the audience this work serves: the non-professional investor who must make sense of the market without the tools available to institutions.

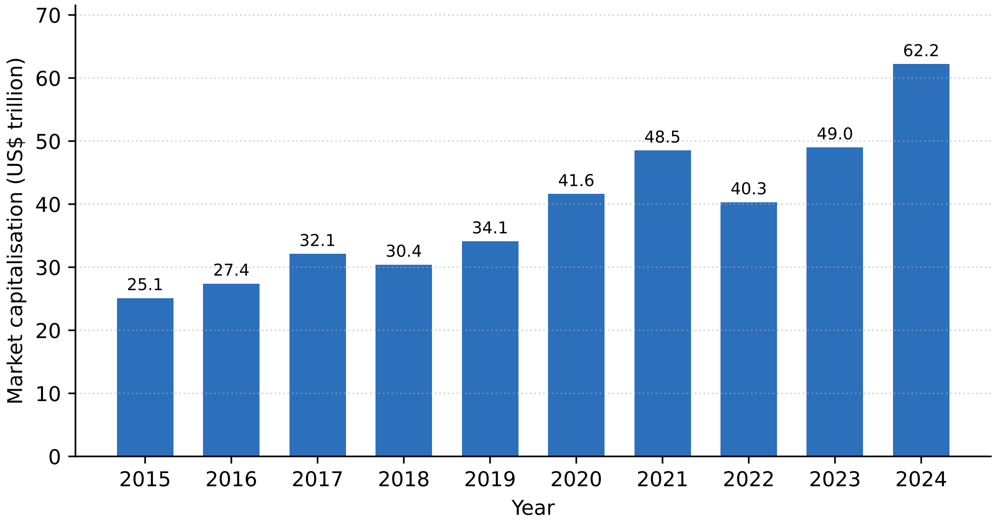

Figure 2.1: US equity market capitalisation, 2015–2024 (US$ trillion). Figure generated from data reported in the SIFMA 2025 Capital Markets Fact Book (Securities Industry and Financial Markets Association (SIFMA) 2025).

2.3 Artificial Intelligence in Finance

AI has been adopted rapidly across financial services. A 2026 industry survey by the Cambridge Centre for Alternative Finance found that 81% of surveyed financial-services firms were adopting AI at some level, 40% of industry respondents reported advanced adoption, and 71% were already using generative AI (Cambridge Centre for Alternative Finance 2026). Typical applications include fraud detection, market analysis and customer service.

This rapid adoption raises a well-documented concern: many high-performing models behave as "black boxes", offering little insight into how a given output was produced. Explainable Artificial Intelligence (XAI) addresses precisely this gap, seeking methods whose reasoning users can follow and trust (Adadi and Berrada 2018; Barredo Arrieta et al. 2020). For a retail investor, who must act on — and bear the consequences of — a system's output, this transparency is not a luxury but a requirement. It is the principle that guides the present work.

2.4 The Information-Overload Problem

A retail investor faces two simultaneous streams of information: continuous price movements across thousands of US tickers, and a constant flow of financial news. Distinguishing a meaningful movement from ordinary volatility, and understanding whether and how a news item is likely to affect an asset, requires time, data and expertise that most non-professionals do not have.

This is the gap the dissertation addresses. Rather than predicting prices, the proposed system detects abrupt movements and relevant news, and — crucially — explains them: what was detected, the reasoning applied, the sources, and concrete historical precedents of analogous events and their observed impact. The remainder of the document develops and evaluates this explainable alerting system.

Chapter 3

Literature Review

This chapter reviews the literature underpinning the system. It covers anomaly detection in financial markets, explainable artificial intelligence, financial natural language processing, and event-study approaches to news–market impact. Each area combines seminal and recent work, and comparative tables summarise the approaches, their mechanisms and their limitations. The chapter closes by positioning the present work and identifying the gap it addresses.

3.1 Anomaly Detection in Financial Markets

Anomaly detection is a mature field with techniques spanning many domains. The survey of Chandola, Banerjee, and Kumar (2009) provides the standard taxonomy — point, contextual and collective anomalies — and groups methods into statistical, distance-based, and machine-learning families. For financial returns, simple statistical methods (deviations from a rolling mean and standard deviation, i.e. a z-score) are attractive because they are transparent and easy to justify, which is an advantage for an explainable system. More expressive machine-learning detectors exist, such as Isolation Forest (Liu, Ting, and Zhou 2008), which isolates anomalies through random partitioning and scales well to large, high-dimensional data; the trade-off is reduced direct interpretability. Table 3.1 compares the two families.

Table 3.1: Anomaly-detection approaches relevant to this work.

Approach	**How it works**	**Strengths / limitations**
Statistical z-score (rolling) (Chandola, Banerjee, and Kumar 2009)	Flags returns that deviate from a rolling mean/standard deviation	Transparent and easy to explain (XAI); assumes locally stable distribution
Isolation Forest (Liu, Ting, and Zhou 2008)	Isolates points via random partitioning (ensemble)	Scalable, handles high dimensionality; less directly interpretable

3.2 Explainable Artificial Intelligence (XAI)

As AI systems are deployed in high-stakes settings, the inability to inspect their reasoning has become a central concern. The surveys of Adadi and Berrada (2018) and Barredo Arrieta et al. (2020) map the field: its motivations, a taxonomy of transparent versus post-hoc methods, and the open challenges toward responsible AI. Among post-hoc, model-agnostic techniques, two are widely used. LIME (Ribeiro, Singh, and Guestrin 2016) explains a single prediction by fitting a simple surrogate model in its local neighbourhood; it is intuitive but its explanations can be unstable and are only locally faithful. SHAP (Lundberg and Lee 2017) attributes a prediction to its input features using Shapley values, offering consistency and a theoretical grounding at a higher computational cost. Table 3.2 summarises these. In this work, explainability is pursued primarily through *transparent design* (statistical detection and retrieved precedents), with feature attribution as an optional complement.

Table 3.2: Explainability approaches considered.

Method	**Type**	**Strengths / limitations**
LIME (Ribeiro, Singh, and Guestrin 2016)	Local surrogate, model-agnostic	Intuitive; explanations can be unstable, only locally faithful
SHAP (Lundberg and Lee 2017)	Shapley-value attribution, model-agnostic	Consistent, theoretically grounded; computationally costly
Transparent design (Barredo Arrieta et al. 2020)	Inherently interpretable logic	No post-hoc approximation; constrains model choice

3.3 Financial NLP and News Representation

Measuring the similarity between news items requires turning text into vectors. Early work learned static word embeddings efficiently with word2vec (Mikolov et al. 2013), but these assign one vector per word, ignoring context. Contextual models based on the Transformer, notably BERT (Devlin et al. 2019), substantially improved language understanding, yet using BERT directly for semantic similarity is expensive because it compares sentence pairs jointly. Sentence-BERT (Reimers and Gurevych 2019) addresses this by producing sentence-level embeddings that can be compared efficiently with cosine similarity — the property this work relies on for precedent retrieval. For the financial domain specifically, FinBERT models fine-tune BERT on financial text, for sentiment (Araci 2019) and for broader financial communications (Yang, Uy, and Huang 2020). Table 3.3 compares these representations.

Table 3.3: Text-representation models for financial news.

Model	**Representation**	**Strengths / limitations**
word2vec (Mikolov et al. 2013)	Static word vectors	Efficient; no context, word-level only
BERT (Devlin et al. 2019)	Contextual token embeddings	Strong understanding; pairwise similarity is costly
Sentence-BERT (Reimers and Gurevych 2019)	Sentence embeddings (siamese)	Efficient cosine similarity; benefits from fine-tuning data
FinBERT (Araci 2019; Yang, Uy, and Huang 2020)	Financial-domain language model	Domain-adapted; oriented to sentiment / communications

3.4 Event Studies and News–Market Correlation

Quantifying the market impact of an event is the province of the *event study*. The methodology of Brown and Warner (1985) established how to measure abnormal returns around an event using daily stock returns, and remains the standard reference for the post-event return windows (e.g. +1, +3 days) used here. Building such an analysis historically requires news aligned to prices; the FNSPID dataset (Dong, Fan, and Peng 2024) provides exactly this — financial news time-aligned with prices for thousands of companies — enabling a news-to-impact knowledge base without scraping. The limitations are the usual ones of any dataset: coverage and time period, discussed in the data card.

3.5 Comparative Analysis and Positioning

Across these areas, the literature offers strong individual components but few integrated, explainable, retail-facing systems that combine them. The choices made in this work, and the alternatives they were weighed against, are summarised in Table 3.4. The guiding principle is *defensible simplicity*: where two options perform comparably, the more transparent and easier-to-explain one is preferred.

Table 3.4: Component choices in this work versus alternatives.

Component	**Chosen**	**Alternative**	**Why**
Anomaly detection	z-score (rolling)	Isolation Forest	Transparency / explainability
News retrieval	Sentence-BERT + cosine	word2vec / raw BERT	Efficient semantic similarity
Impact measurement	Event-study windows	—	Standard, reproducible
Explanation	Rules + precedents (+ SHAP)	LIME-only	Faithful, traceable

3.6 Summary and Research Gaps

The reviewed work supplies mature methods for each piece of the problem: transparent anomaly detection, efficient semantic similarity, domain language models, feature attribution, and event-study impact measurement. What is comparatively scarce is their *integration* into a single, explainable, reproducible system aimed at the retail investor — one that detects events and, for each, retrieves analogous historical news and presents their observed impact as traceable evidence. Addressing this gap, as an engineering contribution, is the purpose of the chapters that follow.

Chapter 4

Methodology

This chapter describes how the system is built. It presents the overall architecture and the two triggers, the two data layers, the method chosen for each component (with its justification), the evaluation design, and the rigour guarantees. Throughout, the guiding principle is *defensible simplicity*: where options perform comparably, the more transparent and explainable one is preferred.

4.1 System Overview and Architecture

The system is organised as a pipeline driven by two independent triggers: an abrupt market movement and an incoming news item. Figure 4.1 shows the components and the flow of data between them. The live layer supplies real-time prices and news; the historical knowledge base, built from FNSPID, supplies the precedents; the anomaly detector and the correlation engine produce the evidence; the explanation engine assembles a traceable account; and the Telegram bot delivers the alert.

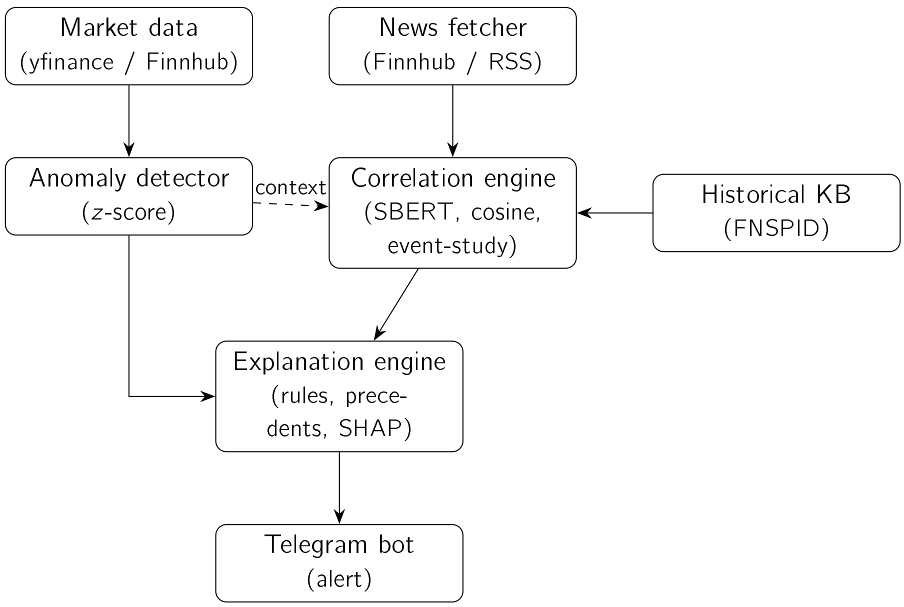

Figure 4.1: System architecture. Solid arrows show the data flow; the dashed arrow shows the anomaly providing context to the correlation engine. The two triggers (market movement via the anomaly detector; news via the correlation engine) converge on the explanation engine.

4.2 Data Architecture: Historical vs. Live

The system uses two clearly separated data layers. The **historical** layer is built offline from FNSPID (Dong, Fan, and Peng 2024), which provides financial news already time-aligned with prices; for each historical news item the system stores its text, date, ticker, a sentence embedding, and the observed post-event price impact. The **live** layer provides, in real time, prices (yfinance, with Finnhub as a fallback) and news (Finnhub and financial RSS feeds). Free news APIs are not used to build history because their free tiers are typically limited to recent windows and are delayed; the exact ticker subset, time window and preprocessing are documented in the data card.

4.3 Statistical Anomaly Detection

Abrupt movements are detected with a transparent statistical rule. Let r_t be the (log-)return of an asset on day t, and let μ_t and σ_t be the mean and standard deviation of returns over a trailing window of w days. The day is flagged as anomalous when the absolute z-score exceeds a threshold k:

$$z_t = \frac{r_t - \mu_t}{\sigma_t}, \qquad \text{anomaly} \iff |z_t| > k. \tag{4.1}$$

This statistical family is standard (Chandola, Banerjee, and Kumar 2009) and is chosen for its transparency: the user can see exactly why an alert fired. More expressive detectors such as Isolation Forest (Liu, Ting, and Zhou 2008) were considered but not adopted as the default, because their reduced interpretability conflicts with the XAI-first goal; the window w and threshold k are documented design choices, evaluated in Chapter 6.

4.4 News–Market Correlation Engine

This is the core of the work. Each news item is encoded into a sentence embedding using a Sentence-BERT model (Reimers and Gurevych 2019), chosen because it allows efficient comparison of texts through cosine similarity. Given a new item, the engine retrieves the k most similar historical news items from the FNSPID-based knowledge base and reports the impact those precedents had. Impact is measured in the manner of an event study (Brown and Warner 1985): the asset's return is computed over fixed post-event windows (e.g. $+1$, $+3$ days). The similarity metric and the impact windows are explicit design choices and form part of the contribution. Crucially, impact is measured strictly after the event, so no future information leaks into the features (Section 4.7).

4.5 Explanation Engine (XAI)

The explanation engine assembles a traceable account for every alert, combining three transparent ingredients: (i) the rule or measure that produced the detection (e.g. the z-score and threshold), (ii) the retrieved historical precedents and their observed impact as evidence, and (iii), optionally, feature attribution over the detector using SHAP (Lundberg and Lee 2017). This combination follows the explainability principles set out in the XAI literature (Adadi and Berrada 2018; Barredo Arrieta et al. 2020): the goal is an explanation the user can follow step by step. Optionally, financial sentiment from FinBERT (Araci 2019) may enrich the explanation; this component is cuttable if it does not add defensible value.

4.6 Evaluation Design

The evaluation is defined in advance, to avoid fishing for favourable results, and is reported in Chapter 6. In summary: the anomaly detector is assessed with precision, recall and F1 against an explicit definition of "true" anomalies, with a baseline and an ablation over window and threshold; the correlation engine is assessed on retrieval quality (precision@k with a relevance rubric, against random and recency baselines) and on the measured event-study impact; and the explanation engine is assessed for fidelity to the actual logic and for usefulness, via a small human-judgement rubric. Because the system performs neither price prediction nor trading, profit-based metrics are explicitly out of scope.

4.7 Methodological Rigour

Three guarantees apply throughout. First, **no lookahead**: features at any instant use only information available at or before that instant, and historical impact uses only post-event windows; this is documented per experiment. Second, **reproducibility**: random seeds are fixed, dependencies are pinned, and every data/results figure is produced by a script in the repository. Third, **honesty**: only results that can be reproduced and explained are reported, and no data, metric or citation is fabricated.

Chapter 5

Implementation

This chapter describes how the system was built. Whereas Chapter 4 justified *what* methods were chosen and *why*, this chapter documents the *engineering*: the environment and tooling, the repository structure and the design principles that keep it testable, the data pipelines, the individual components, and the orchestration of the two end-to-end flows. Consistent with the framing of the work as AI *engineering*, the contribution lies in integrating and applying existing models and tools into a coherent, explainable and reproducible system, not in inventing new algorithms.

5.1 Environment and Tooling

The system is implemented in Python 3.12, chosen for the maturity of its wheels for the machine learning stack. Dependencies are pinned in `requirements.txt` and frozen in a lock file (`requirements.lock.txt`, 72 packages) so the environment is reproducible across machines. The numerical and data stack is `numpy` and `pandas`; the embedding stack is `sentence-transformers` 5.6 on `torch` 2.12 (the CPU build, installed from PyTorch's dedicated index to avoid the much larger CUDA download); figures are produced with `matplotlib`.

Quality is enforced by two always-on gates collected in a single `verify.sh` script: the `pytest` suite and the `ruff` linter (line length 100, with import-order and modernisation rules). Tests that require network access or heavy models are marked and excluded by default — `telegram` (sends a real message) and `sbert` (downloads a model) — so the default run is fast, deterministic and offline, while the marked tests can be invoked explicitly. Secrets live only in a git-ignored `.env` file loaded through `src/config.py`; the session-closing script scans the staged changes for secret-like patterns before every commit. The thesis itself is compiled in continuous integration on every change to `thesis/`, and the PDF is also produced locally by `scripts/build_pdf.sh`.

5.2 Repository Structure and Design Principles

The code is organised as a package per component under `src/`, mirroring the architecture of Figure 4.1; Table 5.1 maps each component to its module and key elements. Three engineering principles recur throughout and are worth highlighting because they are what make the system both testable and defensible.

First, **a thin end-to-end slice was built before breadth**: the market-movement trigger was implemented and proven end to end (price retrieval through to a real Telegram alert) before

Table 5.1: Mapping of components to modules.

Component	Module	Key elements
Configuration	config.py	loads .env; secrets never in code
Market data (live)	market_data/prices.py	get_price_history, log_returns
News fetcher (live)	news_fetcher/fetcher.py	NewsItem; parse_finnhub_news, parse_rss
Anomaly detector	anomaly_detector/detector.py	detect_latest, AnomalyResult
Correlation engine	correlation_engine/	event_study.py, similarity.py
Historical KB	historical_kb/	NewsRecord, Embedder, HistoricalKB
Explanation engine	explanation_engine/explain.py	explain_anomaly, explain_news_impact
Delivery	telegram_bot/sender.py	send_message
Orchestration	main.py	run_thin_slice, run_news_trigger

the heavier correlation engine was added, reducing integration risk. Second, **pure logic is separated from input/output**, and heavy dependencies are imported lazily. Network and model imports (`yfinance`, `requests`, `sentence-transformers`) happen inside the functions that need them, and parsing is split from fetching (for example `parse_finnhub_news` is a pure function tested offline, while `fetch_finnhub_company_news` is a thin HTTP wrapper). As a result the numerical core is unit-tested without a network connection or the machine learning stack. Third, **components are programmed against interfaces**: the embedding step is defined by an `Embedder` protocol with two interchangeable implementations — a dependency-free `HashingEmbedder` (a lexical baseline, also used as the offline test double) and the `SbertEmbedder` (the real Sentence-BERT model). Swapping one for the other changes nothing else, which is exactly what makes the ablation in Chapter 6 a fair comparison.

5.3 Historical Knowledge Base and Data Pipeline

The historical layer is a knowledge base of past news, each entry storing its date, ticker, headline, sentence embedding and measured post-event impact, persisted as JSON Lines (one record per line, human-readable and diff-friendly). The builder, `HistoricalKB.build`, aligns each news item to the market by taking the first trading day *on or after* the news date (via a sorted search over the price index) and measures impact only from that day's close onward; this is the no-lookahead rule made concrete in code, and it also avoids crediting the system with a jump already embedded in the opening price.

The intended primary source is FNSPID (Dong, Fan, and Peng 2024). Its news file was verified to be roughly 23 GB, so `scripts/download_data.py` reads it in a streaming fashion, filtering by ticker and date window chunk by chunk and writing only the selected subset to disk; large data is git-ignored and recreated by the script, while only small samples are versioned. Because downloading and scanning the full corpus is a long, one-off job, the evaluation in Chapter 6 uses an initial knowledge base built from real Finnhub news through the same builder (`scripts/build_kb.py`); the full FNSPID build is identified as future work. Either way the build is identical, differing only in the news source.

5.4 Live Layer: Market Data and News

The live layer reuses the same schema as the historical layer so the two are directly comparable. Market data is obtained from `yfinance`, from which daily log-returns are computed; the import is lazy so that unit tests do not touch the network. News is obtained from the Finnhub *company-news* endpoint and from RSS feeds: `parse_finnhub_news` converts the JSON response into `NewsItem` objects (normalising the epoch timestamp to an ISO date), and `parse_rss` extracts items from a feed using only the standard library. Both were validated against the live Finnhub service. Throughout, the API key is read from the environment and never written to disk or logs.

5.5 Anomaly Detector

The detector implements Equation 4.1 directly. `detect_latest` computes the mean and (sample) standard deviation over the window of days *strictly preceding* the evaluated day, forms the z-score of the latest return, and returns an `AnomalyResult` dataclass carrying the decision together with every quantity that produced it (the z-score, window, threshold, mean and standard deviation). Exposing these values is what allows the explanation engine to be faithful by construction. The evaluation harness applies the same rule across an entire series (`rolling_zscore_flags`) so that the firing-rate and ablation experiments of Chapter 6 use exactly the production logic.

5.6 Correlation Engine

The correlation engine has two pure, vectorised parts. Similarity is computed with cosine over (L2-normalised) embeddings: `top_k_similar` returns the indices and scores of the nearest historical items in a single matrix operation. Impact is computed by `post_event_returns`, which returns the cumulative return at each horizon (+1, +3, +5 days) measured from the event close, with horizons that fall outside the available series reported as `n/a` rather than silently dropped. Given a new item, `HistoricalKB.find_precedents` embeds it with the chosen embedder, ranks the knowledge base by cosine similarity and returns the top precedents with their scores — the evidence later shown to the user.

5.7 Explanation Engine

The explanation engine renders alerts directly from the objects produced upstream, which guarantees that the text never diverges from the computation. For the market trigger, `explain_anomaly` states the move, the z-score, the threshold and window, and the recent mean and standard deviation. For the news trigger, `explain_news_impact` lists the retrieved precedents (date, ticker, similarity score, measured impact and headline) and the average impact across them, and it always ends with an explicit disclaimer that the figures are observed past outcomes, not a prediction — keeping the system within its no-forecasting scope. Financial sentiment (FinBERT (Araci 2019)) remains an optional enrichment and is not required by the current flow.

5.8 Orchestration and Delivery

Two entry points in `main.py` wire the components into the end-to-end flows of Figure 4.1. `run_thin_slice` drives the market trigger — retrieve prices, compute returns, detect, explain and (optionally) send — and `run_news_trigger` drives the news trigger — embed the item, retrieve precedents from the knowledge base, explain and (optionally) send. Delivery is handled by `telegram_bot.send_message`, a thin wrapper over the Telegram Bot API; a real alert was successfully delivered during testing. Periodic scheduling and deployment are deliberately left outside the present scope, as the entry points already encapsulate a complete run.

5.9 Testing and Reproducibility

The system is covered by 40 automated tests (plus the two gated tests above), spanning the anomaly detector, the event study, similarity, the knowledge base, the news parser, the explanation engine and both evaluation modules, together with an offline end-to-end smoke test of the news trigger. The emphasis on pure functions means the great majority of the logic is verified deterministically and offline. All experimental results in Chapter 6 are regenerated by scripts with a fixed seed, and the result figures are written directly into the thesis figure directory, closing the loop between code and document.

Chapter 6

Evaluation

This chapter reports the evaluation of the two core components — the anomaly detector and the correlation/precedents engine — against the protocol defined beforehand in the methodology, together with an end-to-end case study. Following the principle that modest but honest results are preferable to inflated ones, every number reported here is produced by a versioned script with a fixed random seed, and the limitations of each experiment are stated explicitly.

6.1 Experimental Setup

Two real datasets are used. For the anomaly detector, daily closing prices for the fifteen large-capitalisation tickers listed in the data card were retrieved with `yfinance` over a three-year window, yielding 752 daily log-returns per ticker. For the correlation engine, 3,692 real financial-news headlines for the same tickers were collected through the Finnhub *company-news* endpoint — the most recent month available on the free tier — spanning five sectors (technology, banking, energy, health and consumer staples). News headlines and prices are normalised to a common schema so that the live and historical layers are directly comparable. This recent, single-month window is sufficient for the retrieval experiment but is the reason a multi-year corpus (FNSPID) is left as future work for the impact analysis.

Three rules apply throughout, mirroring the evaluation design. First, *no look-ahead*: the z-score of each day uses only the window of strictly earlier days, and event-study impact is measured strictly after the event. Second, *reproducibility*: each result is generated by a script (`scripts/evaluate.py` and `scripts/evaluate_anomaly.py`) with a fixed seed (42), the figures are regenerated from those scripts, and the environment is pinned (Python 3.12 with a dependency lock file). Third, *honesty*: the metrics were chosen before running the experiments, and results are reported as preliminary, with their caveats, rather than tuned for effect.

6.2 Anomaly Detector

The evaluation question is whether the rolling z-score (Chandola, Banerjee, and Kumar 2009) is a better universal detector of abrupt moves than a naïve fixed-percentage threshold. The proxy label for a "true" anomaly is a daily move in the upper percentile of *that ticker*'s own return distribution ($|r| \geq$ the 99th percentile); the baseline is a single global threshold ($|r| \geq 3\%$) applied to every ticker.

Firing-rate consistency (main argument). The strongest evidence is independent of any label. A good universal detector should fire at a comparable rate across instruments; a volatility-blind threshold cannot. Table 6.1 and Figure 6.1 show that the fixed threshold fires on roughly 1% of days for a calm stock (KO) but on about 35% of days for a volatile one (TSLA, NVDA), whereas the z-score fires at a near-constant rate ($\approx$ 2%) for all tickers. The range of firing rates across tickers is twenty times tighter for the z-score, confirming that normalising by recent volatility makes detection comparable across the market.

Table 6.1: Firing rate across the fifteen tickers (yfinance, three years).

Method	Min rate	Max rate	Range
z-score (window 20, ± 3)	0.015	0.032	**0.017**
Fixed threshold ($\lvert r \rvert \geq 3\%$)	0.011	0.354	**0.343**

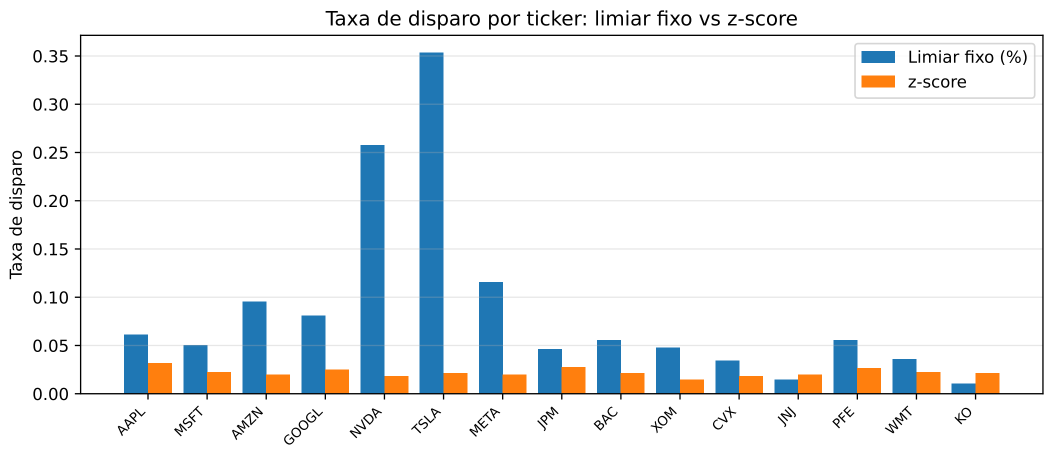

Figure 6.1: Firing rate per ticker. The fixed percentage threshold varies dramatically with each stock's volatility, while the z-score is stable across all tickers.

Agreement with the proxy label (supporting). Against the per-ticker extreme-move label, the z-score attains an F_1 of 0.524 (precision 0.388, recall 0.808) versus 0.216 for the fixed threshold (precision 0.121, recall 1.000): the fixed rule trivially catches every large raw move but at very low precision. A window ablation shows F_1 rising with the estimation window (0.385, 0.524 and 0.687 for 10, 20 and 60 days), indicating that a longer volatility estimate yields a steadier baseline. This label is itself volatility-relative, so some circularity is unavoidable; that is precisely why the firing-rate consistency above — which depends on no label — is treated as the primary evidence.

6.3 Correlation and Precedents Engine

Retrieval quality. The central claim of this work is that semantic retrieval finds genuinely analogous precedents. This is measured as precision@k with a sector proxy, in a *cross-ticker* setting: for each query the k most similar headlines *from other companies* are retrieved, and precision is the fraction sharing the query's sector. Excluding the query's own company prevents the trivial solution of matching on the company name and tests thematic analogy instead. SBERT (Reimers and Gurevych 2019) is compared against a lexical (hashing) baseline and against two trivial baselines — random retrieval (the sector base rate) and recency. Five hundred queries were sampled (seed 42).

Table 6.2: Cross-ticker precision@k by sector (3,692 real headlines, five sectors).

Method	**P@5**	**P@10**
SBERT (semantic)	**0.568**	**0.533**
Lexical baseline (hashing)	0.357	0.328
Recency	0.096	0.077
Random (base rate)	0.245	0.245

As Table 6.2 and Figure 6.2 show, SBERT reaches a precision@5 of 0.568 — about 2.3 times the random base rate (0.245) and well above the lexical baseline (0.357). Recency performs worst, confirming that simply showing the latest news is not a substitute for semantic matching. The lift over both the random and lexical baselines is the honest measure of the value added by the embeddings.

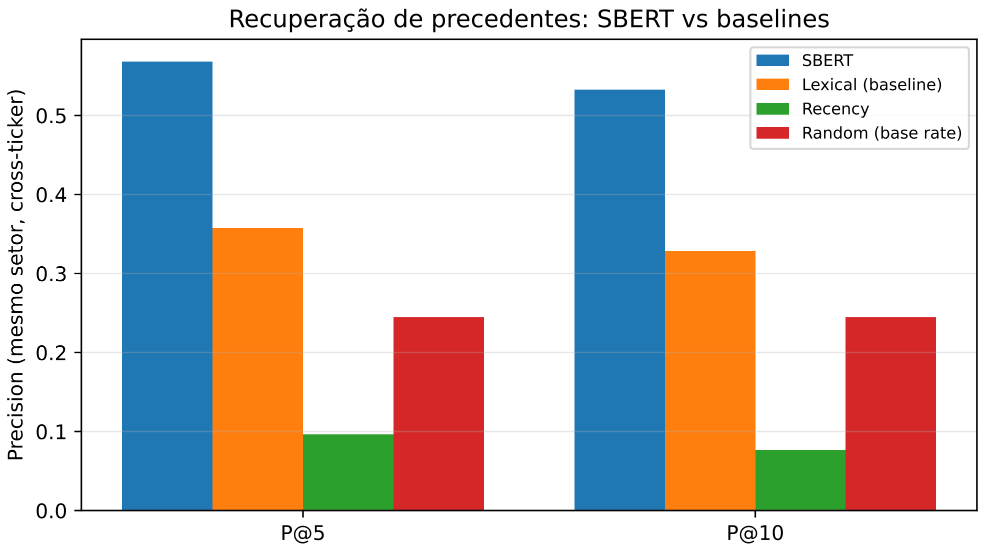

Figure 6.2: Cross-ticker precision@k by sector. SBERT clearly exceeds the lexical, recency and random baselines.

Impact measurement. For each retrieved precedent the engine reports the observed post-event return at +1, +3 and +5 trading days, measured from the event-day close in the style of an event study (Brown and Warner 1985). These figures are observed outcomes presented as evidence, never a prediction (Section 6.6). On the worked knowledge base the measured impacts are coherent with reality — for example, a price-cut headline for one carmaker is followed by a sizable negative three-day return, and a cloud-growth earnings headline by a clear positive one — which provides face validity for the measurement; a rigorous multi-year impact study on the full FNSPID corpus is identified as future work.

6.4 Explanation Quality

Explanation quality is assessed along two axes. *Faithfulness* is guaranteed by construction: the alert text is rendered directly from the same objects used by the system — the computed z-score, window and threshold for the anomaly path, and the exact retrieved precedents with their similarity scores and measured impacts for the news path — so the explanation cannot diverge from the underlying computation. This is verifiable programmatically and is the kind of transparent, design-time explainability advocated in the XAI literature (Barredo Arrieta et al. 2020). *Usefulness* to a retail investor is inherently subjective; a small rubric-based human assessment (clarity, completeness, actionability) on a handful of alerts is planned but has not yet been conducted, and is therefore reported as a limitation rather than a result.

6.5 End-to-End Case Study

To illustrate the news trigger end to end, consider a new headline for NVIDIA, *"Nvidia raises guidance on surging demand for AI data-centre accelerators"*. The system embeds it with SBERT, retrieves the most similar precedents from the knowledge base built from the real news corpus, and renders the alert below (top five precedents, one-day horizon):

```
News alert for NVDA
"Nvidia raises guidance on surging demand for AI data-centre
accelerators"
Potential impact (from 5 similar past events): average 1-day move:
−1.78%
Historical precedents:
- 2026-06-16 META (sim 0.64) +1d −5.44%: "Nvidia Latest Big
Borrower: AI Data Centers..."
- 2026-06-01 BAC (sim 0.63) +1d +1.88%: "Nvidia expands AI platform
with new CPU and PC..."
- 2026-06-18 AMZN (sim 0.61) +1d n/a: "Amazon hopes to challenge
Nvidia by selling AI chips"
- 2026-06-18 AMZN (sim 0.61) +1d n/a: "Amazon's New Weapon: Selling
Custom AI Chips..."
- 2026-06-18 AMZN (sim 0.60) +1d n/a: "Amazon explores selling AI
chips to challenge Nvidia"
Note: precedents are retrieved by semantic similarity; the impact is
the observed past outcome, not a price prediction.
```

Every retrieved precedent is thematically about NVIDIA and AI accelerators, even though the articles were filed under different company feeds (Meta, Bank of America, Amazon). This is direct evidence that retrieval is driven by *meaning* rather than the company name or exact keywords — the very behaviour the correlation engine is designed for. Where a precedent is too recent for its forward window to fall within the available prices, the impact is shown as `n/a`; this is an honest artefact of the one-month free-tier window and further motivates the multi-year corpus. The alert ends with an explicit disclaimer, keeping the system within its no-forecasting scope.

6.6 Discussion

The two experiments support the core design: volatility-normalised detection is consistent across the market where a fixed threshold is not, and semantic retrieval surfaces markedly more sector-analogous precedents than lexical or trivial baselines. Both results rest on real data and are reproducible from versioned scripts.

Several limitations are stated honestly. The sector label is an automatic *proxy* for relevance, not a human judgement; the news corpus is the recent window available from Finnhub rather than the multi-year FNSPID history, and headlines are short, which bounds the semantics that can be captured; the anomaly proxy label is volatility-relative, so the firing-rate argument (label-free) is given primacy; and results use a single seed and no human usefulness study yet. Crucially, and by design, the system performs no price prediction: it does not forecast returns or evaluate trading profit (out of scope). Addressing these points — the full FNSPID corpus, a human relevance and usefulness study, and a multi-year impact analysis — defines the main avenues for further work.

Chapter 7

Conclusion

This chapter summarises the work, revisits the contributions, acknowledges the limitations, and outlines directions for future work.

7.1 Main Conclusions

7.2 Contributions Revisited

7.3 Limitations

7.4 Future Work

Bibliography

Adadi, Amina and Mohammed Berrada (2018). "Peeking Inside the Black-Box: A Survey on Explainable Artificial Intelligence (XAI)". In: *IEEE Access* 6, pp. 52138–52160. doi: 10.1109/ACCESS.2018.2870052.

Araci, Dogu (2019). "FinBERT: Financial Sentiment Analysis with Pre-trained Language Models". In: *arXiv preprint*. arXiv: 1908.10063 [cs.CL].

Barredo Arrieta, Alejandro et al. (2020). "Explainable Artificial Intelligence (XAI): Concepts, Taxonomies, Opportunities and Challenges toward Responsible AI". In: *Information Fusion* 58, pp. 82–115. doi: 10.1016/j.inffus.2019.12.012.

Brown, Stephen J. and Jerold B. Warner (1985). "Using Daily Stock Returns: The Case of Event Studies". In: *Journal of Financial Economics* 14.1, pp. 3–31. doi: 10.1016/0304-405X(85)90042-X.

Cambridge Centre for Alternative Finance (2026). *Global AI in Financial Services Report: Adoption, Impact and Risks*. Cambridge Judge Business School, University of Cambridge. url: https://www.jbs.cam.ac.uk/faculty-research/centres/alternative-finance/publications/2026-global-ai-in-financial-services-report/ (visited on 06/21/2026).

Chandola, Varun, Arindam Banerjee, and Vipin Kumar (2009). "Anomaly Detection: A Survey". In: *ACM Computing Surveys* 41.3, 15:1–15:58. doi: 10.1145/1541880.1541882.

Devlin, Jacob et al. (2019). "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding". In: *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*. arXiv: 1810.04805.

Dong, Zihan, Xinyu Fan, and Zhiyuan Peng (2024). "FNSPID: A Comprehensive Financial News Dataset in Time Series". In: *arXiv preprint*. arXiv: 2402.06698 [q-fin.ST].

Gallup (2025). *What Percentage of Americans Own Stock?* url: https://news.gallup.com/poll/266807/percentage-americans-owns-stock.aspx (visited on 06/21/2026).

Liu, Fei Tony, Kai Ming Ting, and Zhi-Hua Zhou (2008). "Isolation Forest". In: *2008 Eighth IEEE International Conference on Data Mining (ICDM)*, pp. 413–422. doi: 10.1109/ICDM.2008.17.

Lundberg, Scott M. and Su-In Lee (2017). "A Unified Approach to Interpreting Model Predictions". In: *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, pp. 4765–4774. arXiv: 1705.07874.

Mikolov, Tomas et al. (2013). "Efficient Estimation of Word Representations in Vector Space". In: *arXiv preprint*. arXiv: 1301.3781.

Reimers, Nils and Iryna Gurevych (2019). "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks". In: *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pp. 3982–3992. doi: 10.18653/v1/D19-1410.

Ribeiro, Marco Tulio, Sameer Singh, and Carlos Guestrin (2016). ""Why Should I Trust You?": Explaining the Predictions of Any Classifier". In: *Proceedings of the 22nd ACM*

SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1135–1144. doi: 10.1145/2939672.2939778.

Securities Industry and Financial Markets Association (SIFMA) (2025). *2025 Capital Markets Fact Book*. url: https://www.sifma.org/resources/research/fact-book/ (visited on 06/21/2026).

Yang, Yi, Mark Christopher Siy Uy, and Allen Huang (2020). “FinBERT: A Pretrained Language Model for Financial Communications”. In: *arXiv preprint*. arXiv: 2006.08097.

Appendix A

Appendix Title Here

Write your Appendix content here.